Awais Asghar

Hardware Engineer

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Interests

FPGA Design, RTL / SystemVerilog, RISC-V & Computer Architecture, Hardware Accelerators for ML, Hardware-Software Co-Design, Embedded Systems, RTOS, Edge AI

Education

National University of Sciences and Technology (NUST) — Islamabad, Pakistan

Bachelor of Engineering in Electrical Engineering — *Sept 2022 – June 2026*

- CGPA: 3.63/4.0 Specialization GPA: 3.94/4.0 Merit Scholarship, 2024 – 2026
- Senior Design Thesis: ML-Based Hardware Accelerator for Real-Time Image Segmentation on FPGA

Experience

Mitacs Globalink Research Intern — June 2026 – Oct 2026

NRC Herzberg Institute of Astrophysics — *Victoria, BC, Canada*

- Applied ML research on large astrophysical image datasets under Dr. Sebastien Fabbro: model development, HDF5 data pipelines and scientific computing workflows on the CANFAR platform

Chip Design Trainee Engineer — Feb 2025 – May 2026

NUST Chip Design Centre (NCDC) — *Islamabad, Pakistan*

- Designed, simulated and implemented digital systems in SystemVerilog on Xilinx and Intel FPGAs, covering RTL design, verification, timing closure and on-board bring-up under Dr. Hammad M. Cheema
- Built a 5-stage pipelined RV32I RISC-V processor with hazard detection, forwarding and pipeline registers; verified on a Nexys A7 (Artix-7) at 100 MHz with roughly 4× the throughput of the single-cycle design
- Designed a reprogrammable FPGA anti-theft car security system on the DE1-SoC: FSM control, sensor debouncing, siren generation and a fuel-pump safety interlock, verified glitch-free in ModelSim

Deep Learning Research Intern — June 2024 – Aug 2024

Deep Learning Lab (DLL), NUST SEECS — *Islamabad, Pakistan*

- Trained and benchmarked CNN and transfer-learning models with CUDA-accelerated PyTorch and TensorFlow, the model-side groundwork for later FPGA deployment

Embedded Systems Design Research Intern — May 2023 – Aug 2023

ESDAC Lab, NUST — *Islamabad, Pakistan*

- Developed and deployed Embedded Linux on FPGA and DE1-SoC boards under Dr. Usman Zabit: bootloaders, kernel configuration, GPIO bring-up, memory-management tuning and real-time data handling
- Tools: Intel Quartus, Verilog, Linux kernel, RealVNC, FileZilla

Selected Projects

FPGA U-Net Accelerator for Image Segmentation | *SystemVerilog, Vitis HLS, Zybo Z7-20* — 2025 – 2026

- Final year project: hardware-software co-design of a U-Net segmentation accelerator on a Zynq-7000 SoC, from training and INT8 quantization to RTL and bitstream
- Compressed the model 56× to 570 KB so it fits in on-chip BRAM; 97% pixel accuracy and 72% mIoU on the CARLA driving dataset at 1.65 – 2.66 W, with 68/68 RTL testbenches passing

5-Stage Pipelined RISC-V Processor on FPGA | *SystemVerilog, Vivado* | Github Repo — 2025

- IF/ID/EX/MEM/WB pipeline supporting the full RV32I ISA (R, I, S, B, U, J) with a hazard unit, data forwarding and pipeline registers; simulated and tested on hardware at 100 MHz

Single-Cycle RISC-V Processor on FPGA | *SystemVerilog, Vivado* | Github Repo — 2025

- Complete datapath and control: instruction and data memories, ALU, immediate generator, branch comparator and control unit for the whole RV32I instruction set

FPGA-Based Smart Car Security System | *Verilog, Quartus, DE1-SoC* | Github Repo — 2025

- Concealed, reprogrammable anti-theft FSM with debounced sensor inputs, timed siren generation and a fuel-pump interlock; timers and debouncers verified against expected durations

FreeRTOS Smart Energy Monitor | *STM32F746, FreeRTOS, ESP32, C* | Github Repo Dec 2025
- Independent RTOS tasks for ZMPT101B/ACS712 acquisition, Vrms/Irms/kWh computation with DC-offset removal and local display; UART link to an ESP32 that publishes to a Firebase dashboard

Real-Time Fabric Defect Detection on Jetson Nano | *Python, OpenCV, Jetson Nano* | Github Repo Dec 2025
- Classical multi-method pipeline (GLCM, FFT, Gabor, variance, background subtraction, edge-Hough) with IoU box fusion, optimized for live camera input on a low-power edge device; paper in progress

Crowd-Sourced AQI Monitoring System | *Arduino, C++, PMS5003, MQ-series* | Github Repo Nov 2024
- Six-sensor IoT node measuring PM2.5, PM10, CO, temperature, humidity and pressure with real-time AQI estimation at 99% agreement with US Embassy Islamabad readings

Fully Automated Egg Incubator | *ATmega328P, Embedded C, Proteus* | Github Repo June 2024
- Closed-loop temperature and humidity control with stepper-motor egg turning, LCD status and buzzer alarms; 100% pigeon-egg hatch rate in 18 days

Cache Architecture Simulator | *C, C++* | Github Repo Oct 2025
- From-scratch cache simulator replaying memory traces to measure hits, misses and evictions, with an analysis of LRU versus random replacement

Technical Skills

HDL & FPGA: SystemVerilog, Verilog, RTL Design, RISC-V (RV32I), Computer Architecture, Digital System Design, Timing Analysis, Zybo Z7-20 (Zynq-7000), Nexys A7 (Artix-7), DE1-SoC
EDA & Simulation: Xilinx Vivado, Vitis HLS, Intel Quartus, ModelSim, Proteus, NI Multisim, PSpice, LabVIEW, Venus (RISC-V)
Embedded Systems: STM32 (F746), ESP32, Arduino / ATmega328P / ATmega16A, Pixhawk, FreeRTOS, Embedded Linux, Jetson Nano, UART / I2C / SPI, sensor interfacing, PID control
Languages: C, C++, Embedded C, Assembly, Python, CUDA, MATLAB / Simulink
ML for the Edge: PyTorch, TensorFlow, Keras, OpenCV, model quantization (INT8), PYNQ
Tools: Git / GitHub, Linux, IAR Workbench, Atmel Studio, Arduino IDE, Mission Planner / ArduPilot, Jupyter, LaTeX

Publications (In Progress)

Design of a Novel Lightweight U-Net Architecture for Efficient Semantic Segmentation: based on the FPGA accelerator FYP and driving-scene segmentation research at NUST SEECS
Real-Time Fabric Defect Detection on Edge Devices Using Classical Computer Vision: multi-method pipeline deployed on Jetson Nano

Certifications and Relevant Coursework

NUST & NCDC Coursework: Digital System Design, Computer Architecture, RISC-V, Embedded System Design, Digital Signal Processing, Signals and Systems, Electronic Circuit Design, Control Systems, Communication Systems, Electrical Machines, Linux
Deep Learning Specialization (5-Course Series), deeplearning.ai: neural networks, CNNs, sequence models, hyperparameter tuning
Machine Learning Specialization (3-Course Series), deeplearning.ai: supervised and unsupervised learning, reinforcement learning
Deep Learning Course Series, MIT 6.S191: computer vision, generative modeling, transformers

Honors and Achievements

- Mitacs Globalink Research Internship: fully funded international placement at NRC Herzberg, Canada (June to Oct 2026)
- Millennium Fellow: selected among the top 4% globally by the UN Academic Impact and Millennium Campus Network
- NUST SEECS Merit Scholarship, 2024 – 2026
- Dean's Honor List: recognized 4 times for outstanding academic performance
- Prime Minister Youth Laptop Scheme, 2025, awarded for academic excellence
- 3rd place, BISE Multan Pre-Engineering, among 70,000+ students
- Intermediate Scholarship: 100% fee waiver based on exceptional academic performance